Structure of Formal Research Reports

- Abstract. An executive summary of the research. Use about 100 words to summarize your problem, methodology, and major outcomes. Consider adding a list of keywords at the end of the abstract.
- Introduction. **Why** are you conducting this research? Be sure to provide a problem statement, research question(s), and/or hypotheses. What is the general nature of the problem you are addressing? How does it relate to other problems? Why is the problem important to organizations?
- Literature review. **Who** else has conducted research like this? Discuss how other researchers addressed a similar decision problem, what their achievements were, and the advantages and drawbacks of each. Explain how your investigation is similar or different to the state-of-the-art. No need to discuss each individual paper. Instead, identify key characteristics of your topic and review groups of papers relating to these characteristics. Cite relevant papers using Chicago author-date format.
- Methods. **How** are you conducting the research? Discuss the key aspects of your decision analytics modeling, including a statement of your objectives, proposed methodology, model formulation, and the evaluation measure(s) for the performance of the proposed methodology (if applicable). Be sure to discuss your modeling assumptions. If you work on real data, explain your approach to generating the data parameters, and how you verified and validated your model. If you work on theoretical advancement, explain your method with details (put the theory and proofs in the appendix).
- Results. **What** did you learn from the research? Describe how your results apply to solving your real-world decision problem. Be sure to perform some sensitivity analysis of your model parameters.
- Conclusions. **So, what** does it all mean? Review relevant implications for the decision problem at hand. Be sure to conclude your findings, discuss model limitations, and suggest areas for future work. When drawing conclusions, make sure that you relate your study results back to the management problem you have identified.
- References: be sure to cite all references used in the report in Chicago format.
- Appendix/appendices:
    - Additional tables and figures
    - Theorems and proofs (if applicable)
    - Supporting information for verification, validation, or data collection and others
    - Input data files, modeling/programming code, output/listings